Model Selection

Generic Model:

The usage of a simpler model wherever possible:

1. A simpler model is usually more generic than a complex model. This becomes important because generic models are bound to perform better on unseen data sets.
2. A simpler model requires fewer training data points. This becomes extremely important because in many cases, one has to work with limited data points.
3. A simple model is more robust and does not change significantly if the training data points undergo small changes.
4. A simple model may make more errors in the training phase but is bound to outperform complex models when it views new data. This happens because of **overfitting**.
5. Has learnt the core principles of the data
6. Requires fewer training datasets
7.

In the competitive exam analogy, the first person learns using a much more complex mental model than the second one.

Complex OverTrained Model:

1. Very low training error.

2. Does not perform well on new data, especially if the pattern changes
3. High bias
4. Requires large amounts of data to be trained.
5. Has not learnt the core principles of the data
6. Upgrad is over fitting students with repeated, simple data. Likely to not perform well because of lack of understanding of core principles. Should read books more, learn from MIT course, and go from there.

Overfitting Complex Models
Overfitting is a phenomenon wherein a model becomes highly specific to the data on which it is trained and fails to generalise to other unseen data points in a larger domain. A model that has become highly specific to a training data set has 'learnt' not only the hidden patterns in the data but also the noise and the inconsistencies in it. In a typical case of overfitting, a model performs quite well on the training data but fails miserably on the test data.

Bias and Variance

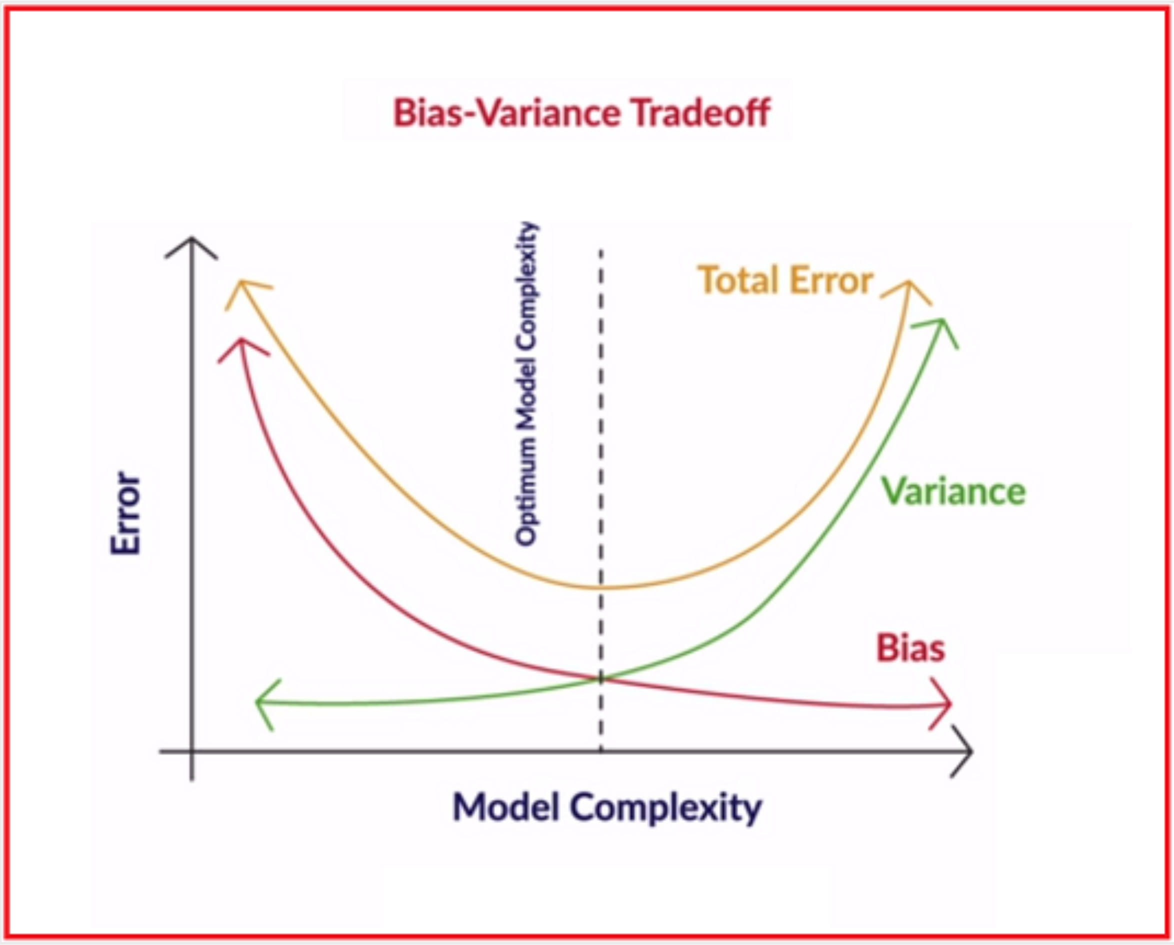
We considered the example of a model memorising the entire training data set. If you change the data set slightly, this model will also need to change drastically. The model is, therefore, **unstable and sensitive to changes in training data**, and this is called **high variance**.

The 'variance' of a model is the **variance in its output** on some test data with respect to the changes in the training data. In other words, variance here refers to the **degree of changes in the model itself** with respect to changes in the training data.

Bias quantifies how **accurate the model is likely to be** on future (test) data. Extremely simple models are likely to fail in predicting complex real-world phenomena. Simplicity has its own disadvantages.

Imagine solving digital image processing problems using simple linear regression when much more complex models such as neural networks are typically successful for such problems. We say that a linear model has a high bias because it is quite simple to be able to learn the complexity involved in the task.

Ideally, we want to reduce both bias and variance because the expected total error of a model is the sum of the errors in bias and variance, as shown in the figure given below.

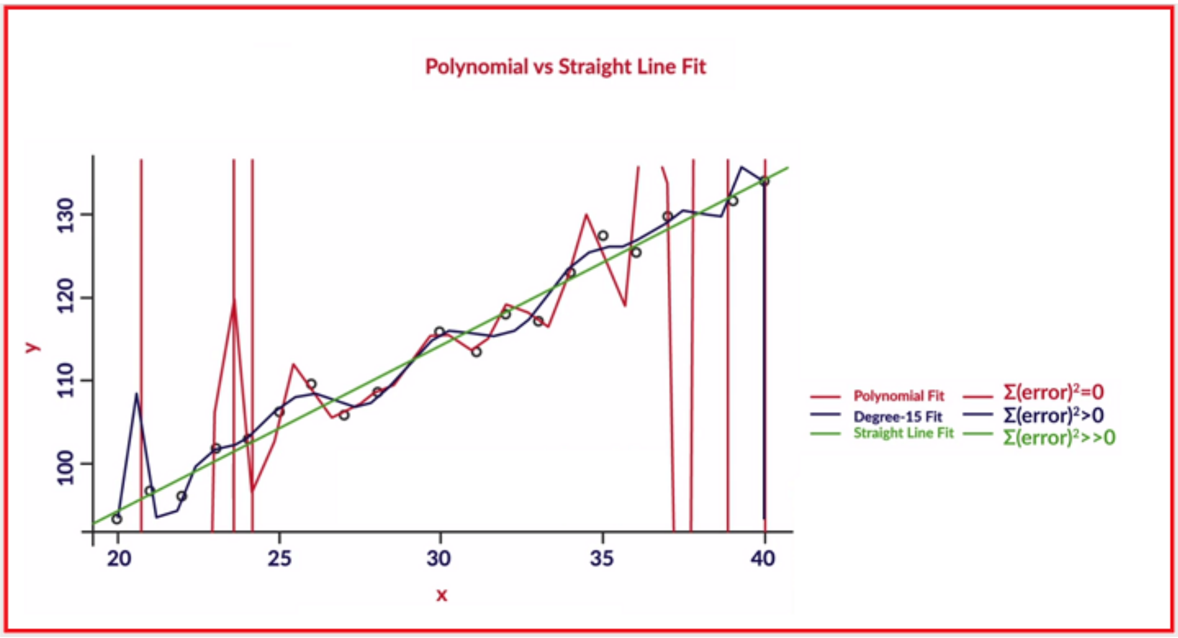


In practice, however, we often cannot have a model with a low bias and a low variance. As the model complexity increases, the bias reduces, whereas the variance increases and, hence, the trade-off.

Recall that in the competitive exam analogy, the first person learns using a much more complex mental model than the second one and has not understood the principles. If on the other hand he can do both, what the heck.

Question:
An artificially generated data set was used to generate data of the form (x, 2x + 55 + e), where e is a normally distributed noise with a mean of zero and variance of 1. The following three regression models have been

created to fit the data: linear, a degree-15 polynomial and a higher degree polynomial that passes through all the training points.



Regularization

Regularization is the process of deliberately simplifying models to achieve the correct balance between keeping the model simple and not too naive. Recall that a few objective ways of measuring simplicity are as follows: choice of simpler functions, fewer model parameters and usage of lower degree polynomials.

Summary

In this session, you learnt about the most fundamental principles of machine learning that you should now be able to apply while building models. The most important points to

re-iterate are as follows:

Occam's Razor

- A model should be as simple as necessary but not simpler than that.
- When in doubt, choose a simpler model.
- Advantages of simplicity are generalisability, robustness, requirement of a few assumptions and less data required for learning

Bias-Variance Tradeoff

- Bias measures how accurately a model can describe the actual task at hand.
- Variance measures how flexible the model is with respect to changes in the training data.
- As complexity increases, bias reduces and variance increases, and we aim to find the optimal point where the total model error is the least.

Overfitting

- A model memorises the data rather than intelligently learning the underlying trends in it.
- This is because it is possible to memorise data, and this is a problem because the real test happens on unseen, real-world data.

//Note: The lecture notes for both sessions of **Model Selection** are provided in the last session of this module.